

Costless Coordination through Public Contracting

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Abstract

A principal incentivizes a team of agents to work on a joint project. This paper proposes a simple mechanism that implements full effort as the unique outcome under any procedure of Iterative Elimination of Weakly Dominated Strategies. The mechanism asks agents to choose between two public messages, “collaborate” and “monopolize,” and the message profile determines their bonuses upon team success. Unlike existing contracts, it ensures full effort at approximately the lowest cost while implementing a non-discriminatory, transparent bonus allocation. Thus, cost efficiency need not come at the cost of fairness or transparency.

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1 Introduction

A trade-off between *cost efficiency*, *fairness*, and *transparency* sits at the heart of team incentives. When workers’ efforts are complementary, each may be willing to work only if others do. To eliminate such coordination failure, existing contracts in the literature must give up at least one of the corners: they either pay a high cost and tolerate discrimination among identical workers (e.g., Winter, 2004), or rely on non-transparent contracts (e.g., Halac, Lipnowski, and Rappoport, 2021). Yet pay transparency laws are spreading (Cullen, 2024). Organizations want open pay structures. The demand for contracts that are transparent, fair, and still deliver effort efficiently is only growing.

This paper shows that such contracts are not only possible but can also be simple. I propose a deterministic mechanism that reconciles these objectives under Iterative Elimination of Weakly Dominated Strategies (IEWDS). Workers publicly choose between a collaborative option and an alternative that grants a “monopoly right.” The design rests on two margins: how much to pay and how to structure the monopoly right. The former makes rejecting the monopoly option a credible signal to teammates that a worker intends to work. The latter addresses uncertainty over who will exercise the monopoly right. Together, these features implement team effort at approximately the lowest cost without sacrificing transparency or fairness. More broadly, the result highlights the potential of IEWDS in contract design: the standard “coordination rent” in public contracting need not arise once we move beyond Nash implementation.

Consider a canonical moral-hazard-in-teams environment with effort complementarities. A team of agents collaborates on a joint project. Each agent privately chooses whether to work or shirk. Working is costly but increases the probability of team success, with a larger effect when more agents exert effort. Only the team outcome is contractible. A principal seeks to incentivize the agents to work in every possible outcome that survives IEWDS.

A natural cost-minimizing and non-discriminatory benchmark contract rewards each agent just enough that he works if and *only if* all others work. Following Ma (1988), I call this contract the second-best benchmark. But these bonuses need not prevent coordination failure: if anyone expects others to shirk, the bonus falls short, and shirking becomes

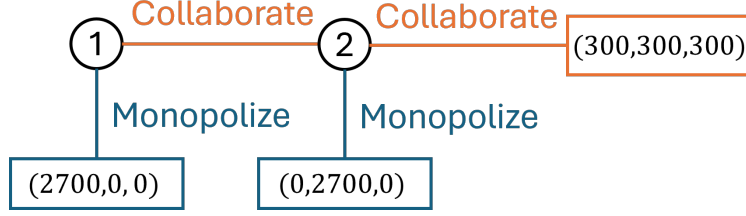


Figure 1: Bonus allocation rule for three agents

Note: This graph illustrates the allocation rule, not the timing of moves: it shows how the mechanism maps a profile of agents’ simultaneous contract choices into a vector of success-contingent bonuses, (b_1, b_2, b_3) . If agents 1 and 2 both choose *collaborate*, each agent receives a bonus of 300 upon team success. If either chooses *monopolize*, the highest-ranked agent who does so receives 2700 upon success, while all others receive nothing. The numbers are illustrative; formal results use bonuses satisfying the strict incentive conditions.

the best response.

The proposed mechanism, which I call *collaborating*, resolves this failure through public contract choices. The principal assigns agents an arbitrary ranking. All but the lowest-ranked agent publicly and simultaneously choose between *collaborate* and *monopolize*. A unanimous choice of *collaborate* leads to a bonus profile arbitrarily close to the second-best. Choosing *monopolize* instead grants the agent a large “monopoly” bonus upon team success and zero to others, but only if the agent is the highest-ranked individual to do so. In this sense, agents can exercise a “monopoly right” in a hierarchical manner. Figure 1 illustrates the contracting procedure for a team of three. After observing the choices, all agents privately decide whether to work.

The mechanism implements a unanimous choice of *collaborate* and full effort as the unique outcome under any IEWDS procedure (Theorem 1). The intuition unfolds in two steps. Take the three-agent case (Figure 1) as an example.

- **Step 1: A unanimous choice of *collaborate* coordinates effort.** The monopoly bonus is large enough that, if an agent chooses *collaborate* but is planning to shirk, his payoff is always no better than, and sometimes worse than, what he would have received by choosing *monopolize* in the first place. As a result, we should expect agents 1 and 2 to work after a unanimous choice of *collaborate*, which makes it optimal for

agent 3 to work as well.

- **Step 2: Coordinated effort makes *collaborate* optimal.** This step proceeds by “upward” induction along the hierarchy. Consider agent 2. Under this hierarchy, his contract choice matters only if agent 1 chooses *collaborate* and thus is expected to work. In that case, if agent 2 also chooses *collaborate*, the resulting unanimous choice of *collaborate* will induce agent 3 to work. The relevant comparison is then simple: agent 2 can either choose *collaborate* while expecting all others to work, or exercise his monopoly right while expecting others to shirk. The monopoly bonus is larger, but less likely to be paid because others shirk. The principal can choose its size so that collaboration still wins for agent 2. A similar argument applies to agent 1.

The mechanism separates the role of hierarchy from the realized pay structure. More precisely, two features distinguish it from the payment hierarchy in Winter (2004). First, the ranking determines whose contractual choice takes precedence, not who earns more in the implemented outcome. Bonuses differ sharply off path, but identical agents receive the same bonus on path. Second, while the mechanism needs a ranking, there is no such thing as the right ranking. Agents can be ranked arbitrarily, even when they differ in effort costs or contributions to team success. The mechanism thus uses a hierarchy of contractual rights to coordinate effort without creating a hierarchy of realized pay.

A related example for which the mechanism may be relevant comes from Buffer, a startup that experimented with a transparent compensation menu. In its early years, Buffer offered employees a choice between roughly 30% more stock options and an additional \$10,000 in annual salary, while publishing its salary formulas and employees’ choices online¹. The choice therefore revealed not only what an employee received, but also what she had willingly given up. Whether this information helps coordinate effort, however, depends on how the menu is designed. The present analysis suggests that, with the right alternatives, choosing a larger stake in the firm’s success over higher fixed pay could credibly signal an intention to work hard. Colleagues who observe that choice may then become more willing to work themselves.

¹See its website for more information: <https://buffer.com/resources/buffer-open-equity-formula/>

Buffer’s arrangement is not a literal instance of the mechanism studied here, as discussed in Section 5. It nevertheless illustrates the broader possibility that public compensation choices can shape workers’ expectations about one another’s effort before work begins.

The remainder of the paper illustrates the mechanism in a two-agent example (Section 2), characterizes the general N -agent result (Sections 3 and 4), and discusses implications for real-world incentive provision (Section 5), including fee-splitting in legal work and equity contracts in startups.

Related Literature

This paper contributes to the literature on contracting against strategic uncertainty in team production. In the setting of public contracting, in which all bonus offers are publicly known, Winter (2004) provides a benchmark analysis. He studies the optimal independent contract where agents cannot influence their own or others’ contracts. The key insight there is that discriminatory bonuses are necessary to ensure full effort in any Nash equilibrium at a minimum cost: a large bonus makes exerting effort a dominant strategy for one agent, so incentivizing the other requires only a small bonus.

Cavounidis and Park (2025) extend this framework by allowing for interdependent contracting, but focus on a constrained class of mechanisms—subcontracting. The principal sets a bonus budget and delegates bonus allocation to one agent. The authors study Nash equilibria in extensive-form-rationalizable strategies (Pearce, 1984) to capture forward induction. Subcontracting can sometimes outperform Winter (2004)’s contract, but not always. In both papers, the constraints on the mechanism space or the equilibrium concept push the cost above the second-best and lead to discrimination. I develop a simple and intuitive mechanism that restores approximately the second-best and eliminates discrimination under IEWDS.²

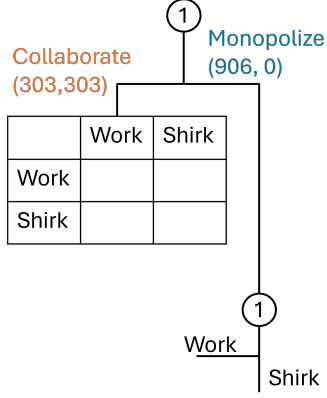
²Iterative (strict) dominance also plays a central role in Winter (2004) and Halac et al. (2021). Wu (2024) compares implementable outcomes under independent contracts across Nash equilibrium, correlated equilibrium, and rationalizability. By contrast, IEWDS and extensive-form rationalizability (EFR) additionally incorporate forward-induction reasoning. The relationship between IEWDS and EFR has been discussed in specific classes of games (Battigalli, 1997; Shimoji, 2002, 2004), but remains unclear in general.

When it comes to private contracting, Halac et al. (2021) find that the optimal private, independent contract outperforms Winter (2004) and eliminates discrimination by creating mutual assurance among agents. Beyond independent contracting, the principal can approximate the second-best with a mechanism involving private messages before or after effort decisions (Ma, 1988; Arya et al., 1997; Cavounidis and Ghosh, 2021) or a private incentive scheme where an agent’s bonus offer depends on the private offers of others (Section 5 of Halac et al. (2021)). Under these mechanisms, agents are not informed of others’ payments, nor can they verify their own contractual terms before work begins. This lack of transparency in one’s own contract is less common in practice. My paper provides a public mechanism that approaches the second-best without asking agents to work under missing contract terms.³

The power of forward induction in selecting equilibria has long been recognized, from early examples in coordination games (Kohlberg and Mertens, 1986; Van Damme, 1989) to applications in pricing and advertising (Glazer and Weiss, 1990; Bagwell and Ramey, 1994), money-burning (Ben-Porath and Dekel, 1992; Hurkens, 1996; Shimoji, 2002), and cheap-talk games (Antić and Persico, 2023). However, existing multi-agent procedures cannot be directly applied to our design problem. They either rely on stronger solution concepts (Hurkens, 1996) or hide parts of agents’ choices (Ben-Porath and Dekel, 1992). Cavounidis and Park (2025) is the first paper that brings forward induction to the team setting. This paper’s contribution is therefore not forward induction itself, but rather a public mechanism that uses this logic to select full-effort, minimum-cost equilibria under full transparency.

Finally, this paper shows how coordination concerns can generate a new form of hierarchy in organizations. Organizational hierarchies are typically tied to differential pay since monetary incentives play a key role in shaping production and allocation decisions (see, e.g., Mookherjee, 2006; Garicano and Van Zandt, 2013; Winter, 2004; Halac et al., 2021; Cavounidis and Park, 2025). In contrast, the mechanism here features a “voice” hierarchy. It assigns

³Various (sequential) mechanisms have also been explored to eliminate undesired equilibria in principal-agent problems with hidden types (Demski and Sappington, 1984; Ma et al., 1988; Mookherjee and Reichelstein, 1990; Glover, 1994) and, more broadly, in contract design problems with externalities (Segal, 2003; Genicot and Ray, 2006; Kapon et al., 2024). Other papers exploit instruments outside contracting to achieve the second-best, such as monitoring (Park and Song, 2025) and an exogenous outside option (Zhao and Zhou, 2026).



dominates *collaborate and then shirk* for agent 1.

2. Message Constraint: $\max\{pb^m - e, p^2b^m\} < b^c - e$. Choosing *monopolize* is strictly worse than *collaborate and then work* for agent 1 when agent 2 chooses *work* after *collaborate*.

The effort constraint guarantees that *collaborate and then shirk* never survives any IEWDS procedure. As a result, agent 2's strategy *shirking after collaborate* will not survive either. Choosing *collaborate* thus induces coordinated effort. The message constraint then rules out any *monopolize* strategy for agent 1. Hence, we must arrive at the unique outcome mentioned above. Since the collaborating bonus $b^c = 303$ exceeds the second-best, $b^c - e = 303 - 200 > pb^c = \frac{1}{3} \times 303$, we can always find a monopoly bonus b^m , say 906, that satisfies both constraints.

The game admits a bad Subgame Perfect Nash Equilibrium in which agent 1 chooses *monopolize and then work*, agent 2 shirks after *monopolize*, and both would shirk after *collaborate*. This strategy profile does not survive IEWDS because for agent 1 *collaborate and then shirk* is dominated by *monopolize and then work*. In this sense, it is irrational for agent 1 to shirk after *collaborate*, and for agent 2 to expect agent 1 to do so in equilibrium.

3 A General Model of Teamwork

A principal aims to incentivize a group of N agents, $i \in \mathcal{N} := \{1, 2, \dots, N\}$, to complete a team project. Each agent privately decides whether to work or shirk, denoted by $a_i \in \{1(\text{work}), 0(\text{shirk})\}$. *Work* incurs a personal cost $e_i > 0$. The team outcome — success or failure — depends on the set of agents who choose to work. For any set of agents, $J \subset \mathcal{N}$, let $P(J)$ denote the probability of team success if the agents in J work and others shirk. $P : 2^{\mathcal{N}} \rightarrow [0, 1]$ satisfies two assumptions: for any $J, J' \subseteq \mathcal{N}$,

1. (monotonicity) if $J \subsetneq J'$, $P(J) < P(J')$;
2. (complementarity) if J, J' are not nested, $P(J \cup J') - P(J) > P(J') - P(J \cap J')$.

The first assumption states that team success becomes more likely as more agents exert effort. The second captures a central source of coordination friction: an agent's marginal

contribution is higher when more teammates also choose to work. By contrast, if efforts were strategic substitutes, coordination problems would not arise.

Only the team outcome is contractible. Agents maximize their expected payment net of any effort cost. If the principal promises a bonus $b_i \in [0, \infty)$ upon team success, agent i 's payoff will be $P(\{j \in \mathcal{N} : a_j = 1\})b_i - e_i a_i$.

The principal's goal is to induce all N agents to work while minimizing expected total payment. To avoid distraction, I do not formally define a general mechanism space or the cost-minimization problem here because the paper focuses on how a particular mechanism approaches the second-best. Instead, I introduce the concept of unique implementation under IEWDS and characterize the second-best benchmark.

Definition 1 (Unique Implementation under IEWDS). *A mechanism uniquely implements full effort and bonus allocation $\mathbf{b} := (b_1, b_2, \dots, b_N) \in [0, \infty)^N$ under IEWDS, if in the reduced normal-form representation of the induced game:*

1. *For every IEWDS procedure, every surviving strategy profile induces the same on-path outcome;⁵ and*
2. *On the path induced by this outcome, each agent i is promised the bonus b_i conditional on team success, and all agents choose to work.*

Definition 2 (Second-Best Benchmark). *The principal's second-best bonus for each agent $i \in \mathcal{N}$ is $\underline{b}_i = \frac{e_i}{P(\mathcal{N}) - P(\{\mathcal{N} \setminus \{i\}\})}$. The second-best expected payment is $P(\mathcal{N}) \sum_{i \in \mathcal{N}} \underline{b}_i$.*

Any admissible finite mechanism that uniquely implements full effort and a bonus allocation $\mathbf{b}$ must satisfy $b_i \geq \underline{b}_i$ for every agent. In particular, it cannot implement an expected total payment strictly below the second-best. Appendix A proves this result and a broader lower bound for mechanisms whose surviving transfer allocations may differ: their worst-case surviving expected payment is at least the second-best level.

⁵We allow dominance by mixed strategies. Under my mechanism, the desired outcome can be reached by an IEWDS procedure using only pure-strategy dominance. Procedures are exhaustive: they stop only when no remaining pure strategy is weakly dominated. Unique implementation here does not require a unique surviving strategy profile; surviving strategies may differ off path.

This benchmark highlights our focus on coordination. If the principal signs a contract with each agent independently and promises each agent slightly more than the second-best bonus $\underline{b}_i$ upon success, miscoordination may happen.⁶ Shirking is a strict best response when all others shirk, while working is a strict best response when all others work. Neither IEWDS nor Nash equilibrium can eliminate the shirking outcome.

4 Collaborating Mechanism

This section starts by describing the *collaborating* mechanism. It can uniquely implement full effort and any bonus allocation satisfying $b_i > \underline{b}_i$ for every agent. By carefully designing how agents “speak” through contract choices, the principal can approach the second-best and implement a non-discriminatory, transparent contract.

4.1 Collaborating Mechanism

The principal selects a target bonus vector $\mathbf{b}^c$ with $b_i^c > \underline{b}_i$ for every $i \in \mathcal{N}$ and ranks the agents arbitrarily. Let $R : \mathcal{N} \rightarrow \mathcal{N}$ be a permutation. Relabel agents by their position in R , so that a lower index means higher priority. Agents $1, \dots, N-1$ are speakers; agent N is the nonspeaker.

Definition 3 (Collaborating Mechanism). *A collaborating mechanism, denoted by $\mathcal{C}(\mathbf{b}^c, \mathbf{b}^m)$, proceeds as follows:*

1. **Contracting stage:** *Each speaker $i < N$ simultaneously sends a public message $m_i \in \{\text{collaborate}, \text{monopolize}\}$.*
 - *If all speakers choose collaborate, each agent receives a success-contingent bonus b_i^c .*
 - *Otherwise, the highest-ranked monopolizing speaker i receives the success-contingent bonus $b_i^m \geq 0$, and all other agents receive zero.*

⁶Winter (2004) (Proposition 4) shows that offering each agent slightly more than the second-best bonus can induce all agents to work in the unique coalition-proof equilibrium. This equilibrium concept assumes a strong cooperative culture in the workplace.

2. **Working stage:** After observing the full message profile, all agents simultaneously choose whether to work or shirk.

Similar to the two-agent example, the monopoly bonus must satisfy two requirements. It must make *monopolize* preferable to *collaborate and then shirk* after unanimous collaboration, but leave full-effort collaboration more attractive than monopolizing alone. For each speaker i , let $G_i := \max\{P(\{i\})b_i^m - e_i, P(\emptyset)b_i^m\}$ be his highest payoff from monopolizing when all other agents shirk. The two terms correspond to working and shirking, respectively. We require

$$P(\mathcal{N} \setminus \{i\})b_i^c < G_i, \quad (\text{EC})$$

$$G_i < P(\mathcal{N})b_i^c - e_i. \quad (\text{MC})$$

The effort constraint (EC) compares the monopoly payoff with the most a speaker can earn by shirking after unanimous collaboration. It makes a suitable monopoly strategy weakly dominate *collaborate and then shirk* at that history. The message constraint (MC) makes full-effort collaboration preferable to monopolizing when all other agents shirk after being monopolized. It is easy to verify that such monopoly bonuses always exist when $b_i^c > \underline{b}_i$ for all i .

Theorem 1. *Given any target bonus vector $\mathbf{b}$ with $b_i > \underline{b}_i$ for all $i \in \mathcal{N}$, the collaborating mechanism $\mathcal{C}(\mathbf{b}^c, \mathbf{b}^m)$ with $\mathbf{b}^c = \mathbf{b}$ and any $\mathbf{b}^m$ satisfying the constraints (EC)-(MC) uniquely implements full effort and bonus allocation $\mathbf{b}$ under IEWDS.*

In the reduced normal form, a speaker's strategy $s_i = (m_i, a_i(m_i, \cdot))$ specifies his message and his effort after every profile of other speakers' messages. It does not specify effort following his own unchosen message. The nonspeaker's strategy is an effort function $s_N = a_N : \{\text{collaborate}, \text{monopolize}\}^{N-1} \rightarrow \{0, 1\}$.

The proof proceeds as follows. We begin with a reference profile that every elimination procedure must preserve. Let $\mathbf{s}^* = (s_i^*)_{i \in \mathcal{N}}$ be defined as follows. Each speaker chooses *collaborate*, works if all other speakers collaborate, and shirks otherwise. The nonspeaker works after unanimous collaboration and shirks otherwise. Lemma 1 establishes that every component of $\mathbf{s}^*$ survives. This step gives us a surviving opponent profile against which the

dominance comparisons in the proof are strict, which helps rule out undesirable strategies. Claim 1 then justifies comparisons with alternatives that may already have been deleted. Lemma 2 shows that zero-bonus agents shirk at surviving profiles. With these observations in place, we prove that unanimous collaboration implies full effort and then use the hierarchy design to eliminate monopoly messages.

Lemma 1. $\mathbf{s}^*$ survives every IEWDS procedure.

Proof of Lemma 1. Suppose by contradiction that the strategy profile $\mathbf{s}^*$ is eliminated under some IEWDS procedure. Then there must exist an agent $i \in \mathcal{N}$ such that s_i^* is weakly dominated by some alternative strategy, while all strategies in $\mathbf{s}_{-i}^*$ remain active. Write the highest collaborating payoff as $V_i = P(\mathcal{N})b_i^c - e_i$.

First, we rule out alternative strategies that are strictly suboptimal against $\mathbf{s}_{-i}^*$. Notice that a strategy choosing *collaborate and then shirk*, or choosing *shirk* after observing all *collaborate* messages for the nonspeaker, earns $P(\mathcal{N} \setminus \{i\})b_i^c < V_i$. A speaker choosing *monopolize* earns at most $G_i < V_i$, since all opponents shirk in $\mathbf{s}_{-i}^*$. Thus the pure strategies attaining the maximum V_i are precisely those choosing *collaborate* (for speakers) and work after all *collaborate* messages. Let E_i denote this set.

Next, we show that against every opponent profile, s_i^* weakly outperforms each strategy in E_i . They agree after all *collaborate* messages. If any other speaker chooses *monopolize*, all strategies in E_i lead to zero bonus for agent i , and s_i^* shirks. Hence, no strategy in E_i can weakly dominate s_i^* in the current round.

Any mixed strategy weakly dominating s_i^* must assign probability one to E_i to attain at least V_i against $\mathbf{s}_{-i}^*$. It therefore cannot improve strictly against any remaining opponent profile. This contradicts the proposed elimination. $\square$

The next observation provides a way to check what strategies cannot survive. Fix an arbitrary IEWDS procedure and let $T = \prod_i T_i$ be the set of surviving profiles after the procedure.

Claim 1. A strategy $s_i \in T_i$ cannot have an alternative strategy s'_i , possibly mixed and possibly previously deleted, that weakly improves its payoff against every $s_{-i} \in T_{-i}$ and strictly improves it against at least one such profile.

Indeed, every alternative strategy s'_i can be replaced by a mixture supported on T_i that weakly outperforms it against T_{-i} . The surviving mixture would then weakly dominate s_i , contradicting the IEWDS definition. Appendix B provides a more detailed justification.

Lemma 2. *In every strategy profile surviving an IEWDS procedure, every agent receiving zero bonus on path shirks.*

Proof of Lemma 2. Suppose $\mathbf{s} \in T$ gives agent i zero bonus while he works. Let $\hat{s}_i$ keep his message, if any, and change only the *work* response at zero-bonus histories to *shirk*. Write $u_i(s)$ for agent i 's expected payoff at the reduced normal-form profile s . Against every opponent profile $\mathbf{s}'_{-i}$, $u_i(\hat{s}_i, \mathbf{s}'_{-i}) \geq u_i(s_i, \mathbf{s}'_{-i})$, whereas at the surviving profile $\mathbf{s}_{-i}$, $u_i(\hat{s}_i, \mathbf{s}_{-i}) = u_i(s_i, \mathbf{s}_{-i}) + e_i$. Claim 1 gives a contradiction, whether or not $\hat{s}_i$ was previously deleted. $\square$

The final part concludes the proof by showing (i) that the costly signal *collaborate* is credible and (ii) how the hierarchy design simplifies each speaker's trade-off to one that resembles the two-agent example.

Proof of Theorem 1. Fix an arbitrary IEWDS procedure and let T be the set of surviving strategy profiles. By Lemma 1, $\mathbf{s}^* \in T$.

Step 1 (Collaborate and then shirk will not survive). Suppose $s_i \in T_i$ chooses *collaborate* and shirks after observing all *collaborate* messages. Consider a monopoly strategy where i chooses *monopolize*, shirks when outranked and otherwise chooses an effort response that attains G_i when everyone else shirks.

Compare the two strategies against any possible $\mathbf{s}_{-i} \in T_{-i}$. If a higher-ranked speaker chooses *monopolize*, the monopoly strategy yields zero and s_i yields at most zero. Now suppose a lower-ranked speaker chooses *monopolize*. The monopoly strategy yields at least $G_i \geq 0$: any additional opponent effort weakly raises the success probability. s_i instead yields at most zero. If all other speakers collaborate, the monopoly strategy again yields at least G_i while s_i leads to at most $P(\mathcal{N} \setminus \{i\})b_i^c < G_i$ by (EC). Thus the monopoly strategy weakly improves against all possible opponent profiles and strictly improves against $\mathbf{s}_{-i}^*$. Claim 1 thus rules out s_i .

Step 2 (The nonspeaker works after unanimous collaboration). Suppose $s_N \in T_N$ prescribes shirking after unanimous collaboration. Consider an alternative that changes only this response to work. Against any surviving opponent profile $s_{-N} \in T_{-N}$, the alternative gives the same payoff unless all speakers collaborate. At that history, Step 1 implies that every speaker works, so the alternative raises the nonspeaker's payoff by $[P(\mathcal{N}) - P(\mathcal{N} \setminus \{N\})]b_N^c - e_N > 0$. In particular, this improvement is strict against the surviving profile s_{-N}^* . Claim 1 therefore rules out s_N , whether or not the alternative was previously deleted.

Step 3 (Speakers collaborate). We use an induction argument here and proceed from speaker $N - 1$ upward. Fix an arbitrary speaker $i < N$. Suppose all surviving strategies of lower-ranked speakers choose *collaborate*; this is vacuous for speaker $N - 1$. We want to prove that all surviving strategies of speaker i also choose *collaborate*. Suppose by contradiction that a surviving strategy $s_i \in T_i$ chooses *monopolize*. We now compare it with s_i^* against any surviving opponent profile $\mathbf{s}_{-i} \in T_{-i}$.

If a higher-ranked speaker chooses *monopolize* in $\mathbf{s}_{-i}$, s_i yields zero by Lemma 2. The alternative s_i^* also yields zero. Otherwise, all lower-ranked speakers choose *collaborate* by the induction hypothesis, so i is the monopolist under s_i . Lemma 2 implies that all his opponents shirk. His payoff is therefore at most G_i . Choosing s_i^* instead produces unanimous collaboration and, by Steps 1 and 2, full effort. It yields $P(\mathcal{N})b_i^c - e_i > G_i$ by (MC). Thus s_i^* is weakly better than the monopoly strategy s_i against all possibly surviving opponent profiles, and strictly better against $\mathbf{s}_{-i}^*$. Claim 1 rules out s_i and completes the induction.

Every profile in T therefore induces unanimous collaboration, full effort, and bonus allocation $b^c = b$. Since the procedure was arbitrary, the result follows. $\square$

Note that $\mathbf{s}^*$ may not be the unique surviving profile. Different surviving strategies may have different effort choices at histories made unreachable by earlier eliminations. In particular, a surviving strategy may choose *work* at such a zero-bonus history, although zero-bonus agents always shirk on path in a surviving profile. These differences do not affect the implemented outcome.

4.2 Efficient Coordination and Discussion

By Theorem 1, for all $\varepsilon > 0$, the collaborating mechanism uniquely implements full effort and the bonus allocation $\mathbf{b} = (b_i + \frac{e_i}{\sum_j e_j} \varepsilon)_{i \in \mathcal{N}}$ under IEWDS. The expected total payment $P(\mathcal{N})(\sum_i b_i + \varepsilon)$ converges to the second-best level as $\varepsilon \rightarrow 0$. The bonus allocation is non-discriminatory: identical agents receive identical bonuses, and the limiting on-path markup,

$$\lim_{\varepsilon \rightarrow 0} \frac{b_i - e_i}{e_i} = \frac{1}{P(\mathcal{N}) - P(\mathcal{N} \setminus \{i\})} - 1,$$

differs across agents only with their marginal contributions to team success. We discuss two key properties of this mechanism below.

The Monopoly Option as a Signaling Device

How can speakers credibly promise to work? They can do so by making an upfront, public, and costly decision: forgoing the monopoly option. This act of signaling makes it possible to coordinate effort under the approximately second-best bonus allocation.

This intuition echoes earlier work (e.g., Kohlberg and Mertens (1986), Van Damme (1989), Glazer and Weiss (1990), and Ben-Porath and Dekel (1992)) on how forward induction selects equilibria in coordination games. To see this, we return to the two-agent example and consider an outside option that directly assigns some payoff to the speaker (see Table 2 for a formal representation). In this game, as long as full effort is a strict Nash equilibrium under our target bonus allocation, we can construct an outside option that offers less than the full-effort outcome, but more than what an agent would receive from unilaterally deviating to shirk. The normal-form game will then have a unique surviving outcome where all agents choose to work under the target bonus allocation.

The specific structure of this outside option and the subsequent effort decisions can be quite flexible. For instance, a “who-to-team-up” variant of the mechanism also works: if any speaker chooses *monopolize*, the highest-ranked among them proceeds alone and decides whether to work or shirk, while the principal excludes all other agents from the project⁷, instead of allowing them to stay and receive zero bonus. As before, the principal can fine-tune the monopoly bonus to approach the second-best.

⁷Exclusion has the same production effect as shirking.

	s_2		
		w	s
s_1	$collaborate, w$	$b_1^c - e_1, b_2^c - e_2$	$pb_1^c - e_1, pb_2^c$
	$collaborate, s$	$pb_1^c, pb_2^c - e_2$	$p^2b_1^c, p^2b_2^c$
	$option$	$z_1, 0$	$z_1, 0$

Table 2: An abstract outside option in the two-agent example

Note: The table presents the reduced normal form game when agent 1 is offered an option that directly pays her $z_1 > 0$ and zero to agent 2. Effort decisions are denoted by $w(ork)$ and $s(hirk)$. $((collaborate, w), w)$ is the unique outcome under IEWDS if $pb_1^c < z_1 < b_1^c - e_1$ and $b_2^c - e_2 > pb_2^c$.

Voice Hierarchy

Extending the two-agent result to a multi-agent setting is initially far from straightforward. When multiple speakers choose *monopolize*, who should be granted that right? Worse yet, when many agents try to signal their intentions simultaneously, coordination can break down. As Ben-Porath and Dekel (1992) show, simultaneous signaling in their game may lead to an outcome in which players burn money without achieving cooperation. While Hurkens (1996) tackles this issue with a stronger solution concept, I change the signaling procedure.

Recall that in our mechanism, agents send their messages simultaneously. The trouble with everyone talking at once motivates a hierarchical structure of our speaking rules. Once an agent speaks, they concede the monopoly right to the next in line. Such a voice hierarchy turns out to prevent the “voice clash” that can arise in simultaneous signaling. The order in which agents speak is quite flexible: unlike Winter (2004) and Cavounidis and Park (2025), it does not matter whose voice ranks higher when agents are asymmetric.

This hierarchical structure lends itself naturally to a sequential variant. Let agents speak in turn, with all messages publicly observed. In particular, if a speaker chooses *monopolize*, the contracting stage ends: he receives his monopoly bonus upon success, and all other agents receive zero. If every speaker chooses *collaborate*, the bonus vector is b^c . All agents then choose effort simultaneously. Figure 3 illustrates the game tree induced by this sequential version, which I conjecture yields the same implementation result.

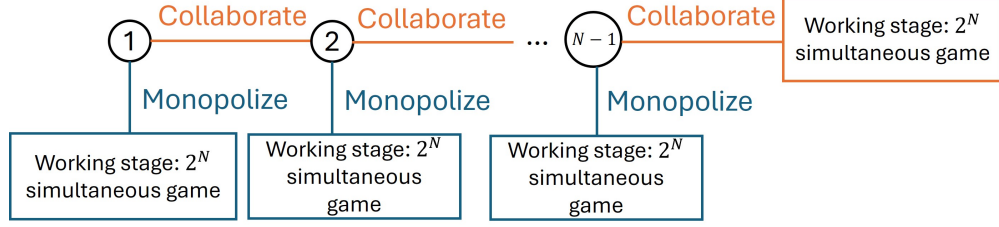


Figure 3: The game tree induced by a sequential collaborating mechanism

While Ben-Porath and Dekel (1992) also see the potential of a sequential signaling procedure, their approach does not support full transparency. Their multi-agent extension requires each agent to leave after sending her signal, so she observes only earlier signals. My approach differs: everyone sees all messages. In my setting, the principal has more flexibility in designing the off-path subgame when the outside option is chosen. It becomes unnecessary to hide lower-ranked or subsequent messages. As a result, in contrast to other second-best mechanisms (e.g., Halac et al., 2021, Cavounidis and Ghosh, 2021), such transparency lets agents confirm their own contractual terms before deciding whether to work.

5 Examples of Applications

This paper introduces a public mechanism that ensures all agents work under the approximately second-best bonus allocation. The mechanism invites agents to make public contract choices that help signal their future efforts before the team project starts. Both the choices and the resulting bonus allocation are public information. Public contracting procedures can therefore help resolve coordination problems in team production without resorting to discriminatory pay. Several real-life examples and applications are discussed below.

Fee Splitting Among Lawyers

Transparent bonus-splitting arrangements are common in legal practice, particularly among lawyers from different firms who collaborate on the same case (Grossbaum, 2022). These collaborations often involve contingent fee matters, where payment depends in part on a successful outcome. Under American Bar Association Model Rule 1.5(e), the client must

agree to the fee division, including each lawyer’s share, and that agreement must be confirmed in writing. Some jurisdictions go further. For certain contingent-fee matters, Florida requires the lawyer assuming primary responsibility to receive at least 75 percent of the fee and limits the secondary lawyer to 25 percent. When lawyers undertake substantially equal work, they may seek court approval for a division giving the secondary lawyer more than 25 percent. These legal requirements help institutionalize transparent, pre-work bonus-splitting contracts, which may help align expectations and reduce coordination failures in team-based legal work.

Equity Contracts in Startups

Startup compensation provides another natural application. Many startups ask employees to trade off higher fixed pay against a larger equity stake in the firm. Buffer, as discussed in the introduction, made this trade-off unusually explicit. Like the *collaborating* mechanism, its arrangement let workers choose publicly between two contract options, revealing what they were willing to forgo. But the arrangements differ in two respects. At Buffer, the alternative to additional stock options was higher fixed pay rather than another success-contingent bonus, and one worker’s contractual terms did not depend on colleagues’ choices.

These differences raise a question: if the principal can pay agents even when the project fails, can independent contracts implement full effort at the same cost? My conjecture is that such an extension would still require agents to observe earlier contract choices before making their own. Contractual terms might then be independent, but the principal may still need the sequence of choices to help agents choose the right contract. Such sequential choice is plausible in practice. Buffer’s public salary spreadsheet, for example, could allow later recruits to observe earlier employees’ salary choices before selecting their own, providing a natural setting for this information structure.

Appendix A Second-Best Benchmark

This section establishes the second-best lower bound for mechanisms whose induced reduced normal-form games have finitely many pure strategies for each agent. The argument uses a

standard result on IEWDS and the existence of a full-effort Nash equilibrium in any terminal game to establish the lower bound.

Mechanisms and implementation. We maintain the assumptions on agents, production technology and effort costs from the main text. Consider mechanisms with a contracting stage followed by a final working stage, at which agents simultaneously choose $a_i \in \{0, 1\}$. Contracting may involve public or private messages, interdependent contractual terms, and mechanism randomization. Effort is non-contractible, so both effort choices remain feasible at every final information set. Let h denote the complete pre-effort history, which need not be public. For $y \in \{0, 1\}$, let $t_i^y(h) \geq 0$ denote agent i 's expected total transfer prescribed at h when the outcome is y , where $y = 1$ denotes team success. The transfer rule depends only on the admissible contracting history and the outcome, not directly on effort.

We assume that the induced reduced normal-form game has finite, nonempty pure-strategy sets S_i and finite expected payoffs. A pure strategy s_i specifies a complete contracting plan and effort choices at every final information set, omitting only behavior at histories ruled out by the agent's own choices. Let $t_i \geq 0$ be agent i 's total realized transfer. For pure or mixed strategy profiles, write

$$u_i(s) = \mathbb{E}_s[t_i - e_i a_i], \quad C(s) = \mathbb{E}_s \left[\sum_i t_i \right].$$

Here expectations include agents' mixing, mechanism randomization, and the production outcome. We denote the principal's second-best expected payment by $C^{SB} := P(\mathcal{N}) \sum_i \underline{b}_i$.

For any procedure D , write its terminal restriction (the set of surviving strategy profiles) as $T^D = \prod_i T_i^D$. We say a mechanism M *implements full effort* if for every procedure D and every surviving profile $s \in T^D$, $a_i = 1$ almost surely for every i . This requirement permits different surviving transfer allocations. Define the worst-case surviving expected payment induced by a mechanism M by

$$C^{\text{wc}}(M) := \max_D \max_{s \in T^D} C(s).$$

Proposition 1. *Suppose an admissible finite mechanism M implements full effort. Then $C^{\text{wc}}(M) \geq C^{SB}$.*

Proof. Fix a procedure D . Since each original pure-strategy set S_i is finite, its terminal restriction is finite and nonempty.⁸ Choose any mixed-strategy Nash equilibrium σ of this terminal game. Such an equilibrium exists by the finite-game Nash theorem and is also a Nash equilibrium of the original game by the equilibrium-preservation property of IEWDS; see Apt (2011, Theorem 5.4(i), arXiv version). Every pure profile in the support of σ belongs to T^D , so all agents work almost surely under σ .

Let H denote the possibly random pre-effort history generated by σ , and define⁹

$$B_i := \mathbb{E}_\sigma[t_i^1(H)], \text{ and } F_i := \mathbb{E}_\sigma[t_i^0(H)].$$

Since everyone works, $\mathbb{E}_\sigma[t_i] = P(\mathcal{N})B_i + (1 - P(\mathcal{N}))F_i$.

Now let agent i keep his contracting behavior unchanged and shirk at every information set at the working stage. If σ_i is mixed, he makes the same draw of a contingent plan and changes only its final effort choices. This is a feasible deviation in the original game. It leaves the pre-effort history distribution unchanged and, because effort choices are simultaneous, leaves every other agent working. Non-contractibility and the absence of subsequent actions mean that the prescribed transfers $t_i^1(h)$ and $t_i^0(h)$ are unchanged. Only the success probability changes. The deviation thus yields $P(\mathcal{N} \setminus \{i\})B_i + (1 - P(\mathcal{N} \setminus \{i\}))F_i$. Since σ is also a Nash equilibrium of the original game, we have

$$P(\mathcal{N})B_i + (1 - P(\mathcal{N}))F_i - e_i \geq P(\mathcal{N} \setminus \{i\})B_i + (1 - P(\mathcal{N} \setminus \{i\}))F_i.$$

Hence $B_i - F_i \geq \underline{b}_i$, and using limited liability,

$$\mathbb{E}_\sigma[t_i] = P(\mathcal{N})(B_i - F_i) + F_i \geq P(\mathcal{N})\underline{b}_i + F_i \geq P(\mathcal{N})\underline{b}_i.$$

Summing over agents yields $C(\sigma) \geq C^{SB}$. Since σ is supported on the terminal restriction,

$$C^{SB} \leq C(\sigma) = \sum_{s \in T^D} \left(\prod_i \sigma_i(s_i) \right) C(s) \leq \max_{s \in T^D} C(s).$$

⁸At any nonempty finite restriction, a pure strategy maximizing the sum of payoffs over the remaining opponent profiles cannot be weakly dominated: a dominating mixture would have a larger sum, although its sum is an average of pure-strategy sums. Thus no round can remove all of an agent's strategies. Finiteness ensures termination.

⁹These expectations average the prescribed transfers over the same distribution of pre-effort histories; they do not condition on a positive-probability success or failure event. So they also apply when $P(\mathcal{N}) = 1$.

The procedure was arbitrary, so we complete the proof. $\square$

The individual bonus bound in the main text follows as a special case. Suppose the mechanism uniquely implements the bonus vector $\mathbf{b}$ as in Definition 1, with zero transfers on failure. In the equilibrium used above, $B_i = b_i$ and $F_i = 0$ for every agent. The inequality $B_i - F_i \geq \underline{b}_i$ therefore gives $b_i \geq \underline{b}_i$ for every i .

Appendix B Justification of Claim 1

To justify the claim, consider any round and fix an available pure strategy. Among mixtures that weakly outperform this strategy against every current pure opponent profile, choose one maximizing the sum of payoffs over those profiles. Such a mixture exists because the feasible set is nonempty and compact.

No pure strategy in its support can be weakly dominated in that round. Otherwise, replacing its positive weight by a dominating mixture would weakly raise every payoff and strictly raise their sum, contradicting maximality. The chosen mixture is therefore supported on strategies that remain after the round.

Starting from any original strategy, apply this replacement whenever a strategy in its support is deleted. There are finitely many rounds, and each replacement weakly improves payoffs against every opponent profile that remains in the terminal game. We thus obtain a mixture supported on T_i that weakly outperforms the original strategy against T_{-i} . The same conclusion holds for an original mixed strategy by applying the argument to each strategy in its support.

Consequently, an alternative that weakly improves on s_i against all of T_{-i} , and strictly improves against one such profile, can be replaced by a mixture supported on T_i with the same properties. This contradicts the definition of the IEWDS procedure.

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